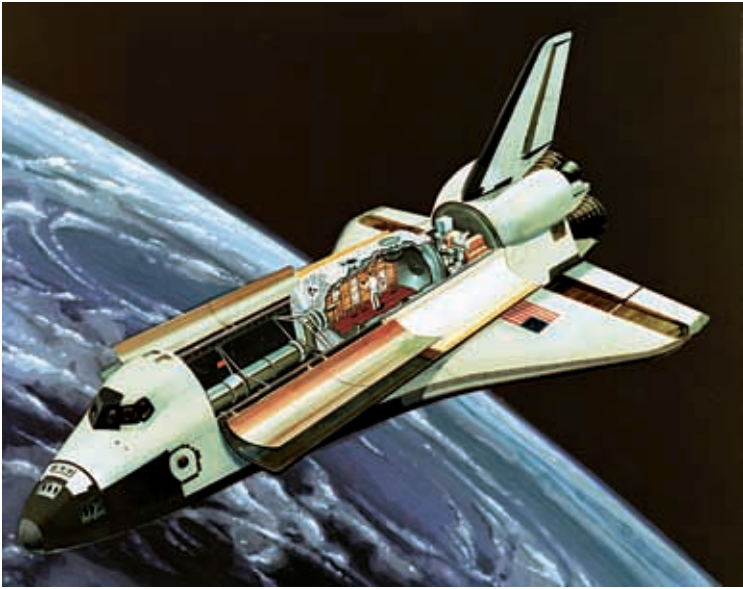




Spacelab was a versatile laboratory carried in the Space Shuttle's cargo bay for special research flights. Its various elements could be combined to accommodate the many types of scientific research that could best be performed in space

support a ton of instruments for each meter of its length – about 3 tons altogether if evenly loaded. With its instruments, the pallet becomes a scientific observatory for studies in astronomy, space physics, and Earth sciences.

Some pallet-mounted instruments are self-contained and automated and require no human intervention during flight. Others need to be switched on and off, guided, or adjusted from controls inside the core module, from the aft flight deck, or remotely from the ground by radio command.

The ESA also made a special component for pallet-only missions. The igloo canister – measuring 10 feet tall, 5 feet in diameter and weighing about 1130 kgs) was mounted beside a pallet, where it held subsystems that supplied power and other utilities to instruments and experiments on the pallet.

The Spacelab system was used on 25 Shuttle flights until 1998, when the laboratories were retired after in-orbit scientific work was transferred to the International Space Station.

Mir

In the 1980s, after the Skylab was retired and Salyut was getting ready to